Reductions for the finite implication E677 $\models_{\text{fin}}$ E255

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Abstract

This note records a proof-oriented package of reductions for the finite implication E677 $\models_{\text{fin}}$ E255 for magmas. The main output is a route graph: the target law at a point is equivalent to a unique right fixed point, to a left-division identity called Lemma L, and to several finite orbit and collision targets. The orbit package now excludes periods 2 and 3, and isolates a period-four gate. The note also records guardrails for older trial routes, including a previously suspected quotient family and a full-shift law that both require a fresh derivation before use.

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1 Introduction

A *magma* is a set M equipped with a binary operation $\diamond$. The two laws studied here are

$$\begin{aligned} \text{E677}: &\qquad x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big), \\ \text{E255}: &\qquad x = ((x \diamond x) \diamond x) \diamond x. \end{aligned}$$

The goal is the finite implication

$$\text{E677} \models_{\text{fin}} \text{E255},$$

that is, whether every finite magma satisfying E677 also satisfies E255. The Equational Theories Project (ETP) paper singles out this finite implication, up to duality, together with the dual implication E2910 $\models_{\text{fin}}$ E47 [1, Problem 8.1]. The ETP blueprint chapter on Equation 677 contains the local structure theory and the standard finite tools used below [4]; the dashboard is a source for cross-checking equation labels [3].

This note is a proof dossier, not a broad search report. The reductions below are organized around a fixed element x of a finite E677-magma. They show that the target law at x is the same as a single right fixed point, that any such fixed-point witness is unique, and that the same target can be expressed as a left-division identity (Lemma L). The later sections turn the remaining cases after the direct fixed-point route into explicit finite orbit and collision targets.

2 Preliminaries and external input

For $a \in M$, write

$$L_a(t) = a \diamond t, \qquad R_a(t) = t \diamond a, \qquad S(t) = t \diamond t.$$

If L_a is bijective, write $a \backslash b$ for the unique element u satisfying $a \diamond u = b$. All products in this note are non-associative and are parenthesized when there is any possible ambiguity.

The finite hypothesis first enters through the following standard ETP fact.

Proposition 1 (Left division in finite E677-magmas)**.** *Let M be a finite magma satisfying* E677*. Then every left translation L_a is bijective. Moreover*

$$a\backslash b = b \diamond ((a \diamond b) \diamond a)$$

for all $a, b \in M$.

Proof. Law E677 gives

$$b = a \diamond \big(b \diamond ((a \diamond b) \diamond a)\big).$$

Thus L_a is surjective. Since M is finite, L_a is bijective, and the displayed preimage formula is exactly the formula for $a\backslash b$. □

Lemma 2 (Division calculus)**.** *Let M be a finite magma satisfying* E677*. Then*

$$a\backslash(a\backslash b) = (a\backslash b) \diamond (b \diamond a), \qquad b\backslash(a\backslash b) = (a \diamond b) \diamond a$$

for all $a, b \in M$.

Proof. For the first identity, Proposition 1, applied to $a\backslash b$ in place of b, gives

$$a\backslash(a\backslash b) = (a\backslash b) \diamond ((a \diamond (a\backslash b)) \diamond a) = (a\backslash b) \diamond (b \diamond a),$$

because $a \diamond (a\backslash b) = b$. For the second identity, Proposition 1 gives

$$b \diamond ((a \diamond b) \diamond a) = a\backslash b,$$

so uniqueness of left division by b gives $b\backslash(a\backslash b) = (a \diamond b) \diamond a$. □

The ETP chapter also records several strategic facts that are used here only as proof discipline. The unrestricted implication E677 $\not\models$ E255 is witnessed by the free E677-magma [4, Section 13.2]; therefore any derivation of the finite implication must use finite structure somewhere. Linear models satisfying E677 satisfy E255, and affine-fiber extensions over such models are ruled out by the ETP structure theory [4, Lemmas 13.3–13.4]. The same chapter gives finite E677-magmas that are not right-cancellative [4, Section 13.1]. Hence no argument below may use right cancellation, associativity, an identity element, or group or quasigroup intuition unless such structure has first been derived. The finite-map lemmas in the ETP finite chapter are a natural source of future proof tools [5, Lemmas 5.5–5.7].

Lemma 3 (Transformed identity)**.** *Let M be a finite magma satisfying* E677*. Then*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)$$

for all $y, z \in M$.

Proof. Substitute $y \diamond z$ for x in E677. This gives

$$y \diamond z = y \diamond \big((y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y)\big).$$

Cancel the common left factor y using Proposition 1. □

Lemma 4 (Key identity)**.** *Let M be a finite magma satisfying* E677*. Then*

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z))$$

for all $y, z \in M$.

Proof. Lemma 3 gives

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y).$$

On the other hand, applying E677 with the parameter $y \diamond z$ gives

$$z = (y \diamond z) \diamond \big(z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z))\big).$$

Left cancellation by $y \diamond z$ gives the identity. $\square$

3 The orbit package

Fix a finite E677-magma M and an element $x \in M$. Define

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of this L_x-orbit, so that $c_d = x$ and $c_0, \dots, c_{d-1}$ are pairwise distinct. Indices on the c_k are read modulo d. Put

$$d_k = c_k \diamond x.$$

Lemma 5 (Orbit recurrence)**.** *For every k,*

$$c_{k-1} = c_k \diamond d_{k+1}.$$

Consequently $d_1 = c_{d-2}$.

Proof. Apply E677 with $y = x$ and the law variable x replaced by c_k:

$$c_k = x \diamond \big(c_k \diamond ((x \diamond c_k) \diamond x)\big) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation gives the recurrence. Taking $k = 0$ gives $c_{d-1} = x \diamond d_1$, so $d_1 = x \backslash c_{d-1} = c_{d-2}$. $\square$

Lemma 6 (Basic left-division identities)**.** *For every k,*

$$x \backslash c_k = c_{k-1}, \qquad c_k \backslash c_{k-1} = d_{k+1}.$$

If

$$p = x \backslash x = c_{d-1}, \qquad q = x \backslash p,$$

then

$$q = d_1 = c_{d-2} = (x \diamond x) \diamond x,$$

and

$$p \diamond (x \diamond x) = d_1, \qquad p \backslash d_1 = x \diamond x.$$

Proof. The first identity follows from $c_k = x \diamond c_{k-1}$. The second is precisely the recurrence in Lemma 5. Since $x \diamond c_{d-1} = c_d = x$, one has $p = c_{d-1}$, and then

$$q = x \backslash p = x \backslash c_{d-1} = c_{d-2} = d_1.$$

Also $d_1 = c_1 \diamond x = (x \diamond x) \diamond x$. Finally, Lemma 2, applied with $a = x$ and $b = x$, gives

$$x \backslash p = p \diamond (x \diamond x).$$

The left side is $q = d_1$, and hence $p \diamond (x \diamond x) = d_1$. The last displayed identity is the definition of left division by p. $\square$

Lemma 7 (A fixed point has the forced index)**.** *If $d_j = x$, then $j \equiv d-2 \pmod{d}$.*

Proof. The condition $d_j = x$ says $c_j \diamond x = x$. Lemma 3 with $y = c_j, z = x$ gives

$$x = x \diamond (x \diamond c_j).$$

On the other hand, E677 with both variables specialized to x gives

$$x = x \diamond (x \diamond d_1).$$

Two left cancellations yield $c_j = d_1 = c_{d-2}$, so the minimality of the orbit period gives the claimed congruence. $\square$

Corollary 8 (No two-cycle in the L_x-orbit)**.** *The period d is never equal to 2. If $d = 1$, then* E255 at x *holds.*

Proof. If $d = 1$, then $x \diamond x = x$, and so

$$((x \diamond x) \diamond x) \diamond x = (x \diamond x) \diamond x = x \diamond x = x.$$

If $d = 2$, Lemma 5 gives $d_1 = c_{d-2} = c_0 = x$. But Lemma 7 applied with $j = 1$ would give $1 \equiv d - 2 \equiv 0 \pmod 2$, a contradiction. $\square$

Lemma 9 (No three-cycle in the L_x-orbit)**.** *The period d is never equal to 3. Consequently any point at which the target has not yet been established must have $d \geq 4$.*

Proof. Assume for contradiction that $d = 3$. Put

$$a = c_1 = x \diamond x, \qquad p = c_2 = x \backslash x, \qquad e = p \diamond x.$$

Since $d_1 = c_{d-2} = c_1$, one has

$$a \diamond x = a. \tag{1}$$

The orbit recurrence gives, for $k = 1$ and $k = 2$,

$$x = a \diamond e, \qquad a = p \diamond a. \tag{2}$$

Apply E677 with law variable a and parameter p. Using (2),

$$a = p \diamond \big(a \diamond ((p \diamond a) \diamond p)\big) = p \diamond (a \diamond (a \diamond p)).$$

Comparing this with $a = p \diamond a$ and left-cancelling by p gives

$$a \diamond (a \diamond p) = a.$$

Comparing with (1) and left-cancelling by a gives

$$a \diamond p = x. \tag{3}$$

Together with $x = a \diamond e$, this implies $e = p$ by left cancellation by a.

Now apply E677 with law variable x and parameter a. Using (1),

$$x = a \diamond \big(x \diamond ((a \diamond x) \diamond a)\big) = a \diamond (x \diamond (a \diamond a)).$$

Comparing with $x = a \diamond e$ and left-cancelling by a gives

$$x \diamond (a \diamond a) = e = p.$$

Since $x \diamond a = p$, left cancellation by x gives

$$a \diamond a = a. \tag{4}$$

Finally, Lemma 3 with $y = a$ and $z = x$, together with (1) and (4), gives

$$x = (a \diamond x) \diamond ((a \diamond (a \diamond x)) \diamond a) = a \diamond ((a \diamond a) \diamond a) = a.$$

This contradicts the minimality of the period, since $c_0 \neq c_1$ when $d = 3$. □

Lemma 10 (Period-four gate). *Assume that the L_x-orbit of x has period 4. Put*

$$a = c_1 = x \diamond x, \qquad q = c_2, \qquad p = c_3 = x \backslash x, \qquad r = q \diamond x.$$

Then

$$x \diamond a = q, \qquad a \diamond x = q, \qquad a \diamond r = x, \qquad x \diamond q = p, \qquad p \diamond a = q.$$

Moreover $r \notin \{x, a, q\}$. Thus, with respect to this element r, a period-four proof has two remaining targets: eliminate either $r = p$, or the case where r lies outside the principal L_x-orbit.

Proof. The equality $x \diamond a = q$ is the definition of c_2, and $a \diamond x = q$ is $d_1 = c_{d-2}$ in the case $d = 4$. The remaining displayed relations are the orbit definition and the orbit recurrence at $k = 1$ and $k = 3$:

$$a \diamond r = x, \qquad p \diamond a = q.$$

If $r = x$, then $x = a \diamond r = a \diamond x = q$, contradicting $c_0 \neq c_2$. If $r = a$, then $q \diamond x = a$ and $a \diamond a = x$. Lemma 3, applied with $y = q$ and $z = x$, gives

$$x = (q \diamond x) \diamond ((q \diamond (q \diamond x)) \diamond q) = a \diamond ((q \diamond a) \diamond q).$$

Comparing with $x = a \diamond a$, left cancellation by a gives

$$(q \diamond a) \diamond q = a. \tag{5}$$

Lemma 4, applied with $y = q$ and $z = x$, gives

$$(q \diamond (q \diamond x)) \diamond q = x \diamond (((q \diamond x) \diamond x) \diamond (q \diamond x)).$$

Using $q \diamond x = a$, $a \diamond x = q$, and (5), this becomes

$$a = x \diamond (q \diamond a).$$

Since $a = x \diamond x$, left cancellation by x gives $q \diamond a = x$. Then (5) gives $a = x \diamond q = p$, contradicting $c_1 \neq c_3$.

If $r = q$, then $q \diamond x = q$ and $a \diamond q = x$. Lemma 4, applied with $y = a$ and $z = x$, gives

$$(a \diamond (a \diamond x)) \diamond a = x \diamond (((a \diamond x) \diamond x) \diamond (a \diamond x)).$$

Using $a \diamond x = q$, $q \diamond x = q$, and $a \diamond q = x$, this becomes

$$q = x \diamond (q \diamond q).$$

Since $q = x \diamond a$, left cancellation by x gives $q \diamond q = a$. Applying Lemma 4 with $y = q$ and $z = x$, and using $q \diamond x = q$, now gives

$$x = (q \diamond (q \diamond x)) \diamond q = x \diamond (((q \diamond x) \diamond x) \diamond (q \diamond x)) = x \diamond (q \diamond q) = q,$$

contradicting $c_0 \neq c_2$. □

Lemma 11 (Period-four branch bookkeeping). *Assume that the L_x-orbit of x has period 4, with notation as in Lemma 10. Put*

$$s = p \diamond x, \qquad h = r \diamond q.$$

Then

$$q \diamond s = a, \qquad (p \diamond q) \diamond p = s, \qquad a \diamond h = r.$$

If moreover $r \notin \{x, a, q, p\}$, and if

$$t = q \diamond a,$$

then

$$x \diamond t = r, \qquad t \notin \{x, a, q, p, r\}.$$

Consequently a period-four external counterexample has at least two elements outside the principal L_x-orbit, namely r and t.

Proof. The equality $q \diamond s = a$ is the orbit recurrence at $k = 2$, since $s = d_3 = p \diamond x$. Lemma 4, applied with $y = x$ and $z = q$, gives

$$(x \diamond (x \diamond q)) \diamond x = q \diamond (((x \diamond q) \diamond q) \diamond (x \diamond q)).$$

Using $x \diamond q = p$ and $x \diamond p = x$, this becomes

$$a = q \diamond ((p \diamond q) \diamond p).$$

Since the orbit recurrence also gives $a = q \diamond s$, left cancellation by q gives $(p \diamond q) \diamond p = s$.

Next apply E677 with law variable r and parameter a. Using $a \diamond r = x$ and $x \diamond a = q$, one obtains

$$r = a \diamond \big(r \diamond ((a \diamond r) \diamond a)\big) = a \diamond (r \diamond (x \diamond a)) = a \diamond (r \diamond q) = a \diamond h.$$

Finally assume $r \notin \{x, a, q, p\}$. Lemma 3, applied with $y = a$ and $z = r$, gives

$$r = (a \diamond r) \diamond ((a \diamond (a \diamond r)) \diamond a) = x \diamond ((a \diamond x) \diamond a) = x \diamond (q \diamond a) = x \diamond t.$$

Since L_x permutes the principal orbit cyclically, no element of $\{x, a, q, p\}$ can be mapped by L_x to the external element r. Hence $t \notin \{x, a, q, p\}$. Also $t \neq r$, for otherwise $q \diamond a = q \diamond x$, and left cancellation by q would give $a = x$, contrary to the period-four orbit. This proves the claim. $\square$

4 The route graph

The following display is the dependency graph used in the rest of the note. The first row contains equivalent fixed-point targets. The two lower branches describe two proof mechanisms available after the direct fixed-point route: collision lifting and right-collision rigidity.

$$\begin{array}{ccccccc}
\text{E255 at } x & \Longleftrightarrow & q \diamond x = x & \Longleftrightarrow & \exists u\ (u \diamond x = x) & \Longleftrightarrow & q\backslash x = x \\
 & & \Updownarrow & & & & \Updownarrow \\
 & & \text{unique witness } q & & & & ((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x \\
 & & & & & & \Updownarrow \\
 & & & & & & (q \diamond x) \diamond q = p.
\end{array}$$

When working under the contrary assumption $q \diamond x \neq x$, two concrete branches are

$$\begin{array}{ccccc}
d_i = d_j = e & \Longrightarrow & (c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j & \dashrightarrow & \text{finite counting pressure on } y \mapsto y \diamond e, \\
x \notin \operatorname{im} R_x & \Longrightarrow & y \diamond x = z \diamond x = r & \Longrightarrow & (y \diamond r) \diamond y = r\backslash x.
\end{array}$$

The dashed arrow marks a lemma to supply: the collision branch is most useful as a finite-counting problem rather than a direct propagation problem.

Target 12 (Single local gap)**.** For fixed x, put $a = x \diamond x$, $q = a \diamond x$, and $p = x\backslash x$. The local target is any one of the equivalent statements

$$q \diamond x = x, \qquad ((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x, \qquad (q \diamond x) \diamond q = p.$$

Under the contrary assumption $q \diamond x \neq x$, the right-translation argument gives a fiber

$$F_r = \{t : t \diamond x = r\}, \qquad A_r(t) = t \diamond r,$$

with at least two elements on which A_r is injective, and with the rectangle law

$$(t \diamond r) \diamond t = r\backslash x \qquad (t \in F_r).$$

The target in this branch is to turn this injective splitter into a finite contradiction, a fixed point, or Lemma L. The splitter is useful structure to exploit; the next lemma should add a small-image, collision, or fixed-point mechanism.

5 The fixed-point route

The fixed-point route is sharper than the existential formulation suggests.

Proposition 13 (Unique right fixed-point witness)**.** *Let M be a finite* E677*-magma and fix $x \in M$. If $u \diamond x = x$, then*

$$u = q = (x \diamond x) \diamond x.$$

Consequently E255 at x *is equivalent to the existence of u with $u \diamond x = x$.*

Proof. The first half of the proof is the argument already used in Lemma 7. If $u \diamond x = x$, then Lemma 3 gives

$$x = x \diamond (x \diamond u).$$

The specialization $y = x$, $x = x$ in E677 gives

$$x = x \diamond (x \diamond q).$$

Two left cancellations give $u = q$. Since $q = (x \diamond x) \diamond x$, the identity $q \diamond x = x$ is exactly E255 at x. Thus E255 at x is equivalent to the existence of a right fixed point over x, and the witness is forced to be q. □

Proposition 14 (Equivalent local targets)**.** *With the notation above, the following are equivalent:*

1. E255 at x, *equivalently $q \diamond x = x$;*
2. $q\backslash x = x$*;*
3. $((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x$*;*
4. $(q \diamond x) \diamond q = p$.

Proof. The equivalence of (1) and (2) is just uniqueness of left division by q. To identify the left quotient, apply Lemma 3 with $y = x \diamond x$ and $z = x$:

$$x = q \diamond \big(((x \diamond x) \diamond q) \diamond (x \diamond x)\big).$$

Therefore

$$q\backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x),$$

which proves the equivalence of (2) and (3), because $q = (x \diamond x) \diamond x$.

For (4), use Proposition 1:

$$q\backslash x = x \diamond ((q \diamond x) \diamond q).$$

Since $p = x\backslash x$, injectivity of L_x gives

$$x \diamond ((q \diamond x) \diamond q) = x \quad \Longleftrightarrow \quad x \diamond ((q \diamond x) \diamond q) = x \diamond p \quad \Longleftrightarrow \quad (q \diamond x) \diamond q = p.$$

Together with (1)$\Longleftrightarrow$(2), this proves the claim. □

Definition 15 (Lemma L)**.** For the fixed element x, write $\mathrm{L}(x)$ for the identity

$$((x \diamond x) \diamond ((x \diamond x) \diamond x)) \diamond (x \diamond x) = x.$$

By Proposition 14, $\mathrm{L}(x)$ is equivalent, in the present finite E677 setting, to E255 at x.

Remark 16. Lemma L is a finite proof target. Because the unrestricted implication E677 $\models$ E255 is false [4, Section 13.2], this route should make its finite step visible: for example through left bijection, orbit periodicity, a finite-map lemma, or a collision argument. A derivation that rewrites E677 into Lemma L should identify exactly where finite structure enters.

Lemma 17 (Fixed-map and edge-product form)**.** *For the fixed element x, define*

$$H_x(t) = ((x \diamond t) \diamond x), \qquad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x\backslash t = t \diamond H_x(t) \tag{E}$$

for every $t \in M$, and

$$F_x(L_x(t)) = L_x(H_x(t)). \tag{Conj}$$

Consequently F_x and H_x have the same finite cycle structure. Moreover E255 at x *is equivalent to either of these maps having a fixed point. When fixed points exist, they are unique: the fixed point of F_x is*

$$x\backslash q,$$

and the fixed point of H_x is

$$x\backslash(x\backslash q).$$

Thus, under the contrary assumption $q \diamond x \neq x$, both maps are fixed-point-free and every cycle of either map has length at least 2.

Proof. Apply E677 with parameter x and law variable t:

$$t = x \diamond \big(t \diamond ((x \diamond t) \diamond x)\big) = x \diamond (t \diamond H_x(t)).$$

This is exactly (E). The conjugacy identity is immediate:

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = L_x(H_x(t)).$$

Since L_x is bijective by Proposition 1, (Conj) gives the same cycle structure: L_x carries each H_x-cycle to an F_x-cycle, and L_x^{-1} carries each F_x-cycle back to an H_x-cycle.

If $F_x(z) = z$, then

$$x \diamond (z \diamond x) = z.$$

The left-inverse formula for L_x, applied to z, also gives

$$x \diamond \big(z \diamond ((x \diamond z) \diamond x)\big) = z.$$

Left cancellation by x, then by z, yields $(x \diamond z) \diamond x = x$. Hence a right fixed-point witness over x exists, and Proposition 13 gives E255 at x. More precisely, the same proposition gives

$$x \diamond z = q,$$

so any fixed point of F_x must be $z = x\backslash q$. Conversely, if $u \diamond x = x$, set

$$z = u \diamond ((x \diamond u) \diamond x).$$

Then $x \diamond z = u$ by E677 with parameter x, and another use of the same identity gives $z = x \diamond (z \diamond x)$, so $F_x(z) = z$. Taking the forced witness $u = q$, this fixed point is exactly $x\backslash q$. Finally, (Conj) shows that t is fixed by H_x if and only if $L_x(t)$ is fixed by F_x. Therefore the corresponding fixed point of H_x is $x\backslash(x\backslash q)$. □

Remark 18. Identity (E) turns the fixed-point route into a finite dynamics problem. If $H_x(t) = t$, then $x\backslash t = t \diamond t$, equivalently $L_x(S(t)) = t$, which is the last fixed-point form in the ETP Lemma 13.2 list. At a point where the fixed-point target has not yet been established, the forward orbit of any element under H_x must eventually enter a cycle of length greater than 1. A finite proof via this route would have to show that such an H_x-cycle is incompatible with the edge products $x\backslash t = t \diamond H_x(t)$.

Remark 19 (Cycle-target calibration). The previous remark is a local finite-dynamics hint, not a global no-cycle target. In the standard linear model on $\mathbb{F}_5$,

$$u \diamond v = 2u - v,$$

one checks directly that E677 and E255 both hold. For fixed x,

$$H_x(t) = ((x \diamond t) \diamond x) = 3x - 2t.$$

Thus H_x fixes x, while on the other four elements it has a 4-cycle; for example, when $x = 0$, the cycle is

$$1 \mapsto 3 \mapsto 4 \mapsto 2 \mapsto 1.$$

So the finite-map target is the sharper local one: rule out the fixed-point-free H_x-dynamics forced by the contrary assumption, using the edge products $x\backslash t = t \diamond H_x(t)$ or an actual finite-map identity of the kind used in the ETP finite-map lemmas.

Remark 20 (Finite-map lemmas need a non-tautological input). The finite-map lemmas from the ETP finite chapter apply only after one has produced an identity of the form $f = f \circ f \circ g$ or $f = g \circ f \circ f$ for concrete self-maps of a finite set. The immediate identities

$$L_x = L_x \circ L_x \circ L_x^{-1}, \qquad L_x^{-1} = L_x^{-1} \circ L_x^{-1} \circ L_x$$

only recover the tautology $L_x = L_x \circ L_x^{-1} \circ L_x$. They do not involve the right translation R_x, the maps $H_x = R_x \circ L_x$, $F_x = L_x \circ R_x$, or the gate $t \mapsto (t \diamond x) \diamond t$. A useful finite-map proof will display a non-tautological identity for one of those maps before invoking Lemma 5.5 or Lemma 5.6.

6 Collision target

The orbit recurrence suggests a collision strategy, but the exact missing step is narrow.

Proposition 21 (Collision lift)**.** *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j. \tag{C}$$

If in addition $i \not\equiv j \pmod{d}$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Lemma 3 with $y = c_k$ and $z = x$ gives

$$x = d_k \diamond ((c_k \diamond d_k) \diamond c_k).$$

If $d_i = d_j = e$, left cancellation by e yields (C). If also $c_i \diamond e = c_j \diamond e = s$, then (C) becomes $s \diamond c_i = s \diamond c_j$. Left cancellation by s gives $c_i = c_j$, contradicting the minimal period unless $i \equiv j \pmod{d}$. ☐

Target 22 (Collision-counting target)**.** Turn a nontrivial collision $d_i = d_j = e$ into a finite counting contradiction, or into the fixed point $q \diamond x = x$. More precisely, let

$$F_e = \{y \in M : y \diamond x = e\}, \qquad A_e(y) = y \diamond e.$$

Proposition 21, and more generally Proposition 23, show that A_e is injective on F_e. Thus the old direct target $c_i \diamond e = c_j \diamond e$ is not a plausible standalone propagation lemma for a genuine collision: it is precisely what the splitter forbids unless a separate argument has already produced the fixed point. A collision proof should instead find a natural finite set B_e such that

$$A_e(F_e) \subseteq B_e$$

and $|B_e| < |F_e|$, unless $q \diamond x = x$.

The common shortcut "prove $k \mapsto d_k$ injective" is not by itself a proof of the main theorem. It becomes useful only if the argument also connects the resulting injectivity to the forced fixed point $q \diamond x = x$, or if it supplies the counting pressure requested in Target 22.

7 Right-collision constraints

Work under the contrary assumption $q \diamond x \neq x$. Proposition 13 says that no element u has $u \diamond x = x$. Thus $x \notin \operatorname{im} R_x$. Since M is finite, R_x is not injective, and there exist $y \neq z$ and $r \neq x$ such that

$$y \diamond x = z \diamond x = r.$$

The following proposition records the forced shape of such a right collision.

Proposition 23 (Right-collision rectangles)**.** *Let $y \diamond x = r$. Then*

$$(y \diamond r) \diamond y = r \backslash x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Lemma 3 with $y = y$ and $z = x$ gives

$$x = r \diamond ((y \diamond r) \diamond y),$$

which is exactly (R). If $y \diamond r = z \diamond r = s$, then (R) gives $s \diamond y = s \diamond z$. Left cancellation by s gives $y = z$. $\square$

This proposition is used here as a proof constraint. If the fixed-point route has not yet produced $q \diamond x = x$, then right-collision fibers must split under the second probe $t \mapsto t \diamond r$, while all elements in the fiber preserve the same twisted diagonal value $(t \diamond r) \diamond t = r \backslash x$. A proof may use this rigidity for counting or collision lifting.

For bookkeeping, it is often useful to write the left translations as labels. Since all left translations are permutations, one may write

$$a \diamond b = \sigma(a)(b)$$

with $\sigma(a) \in \mathrm{Sym}(M)$. Law E677 is then the labeled- permutation condition

$$\sigma(y)\big(\sigma(x)(\sigma(\sigma(y)(x))(y))\big) = x.$$

The labeling condition is not merely a group-theoretic condition on a subgroup of $\mathrm{Sym}(M)$; the labeling map $a \mapsto \sigma(a)$ is part of the data. In fact this labeling map is injective. If $\sigma(y) = \sigma(z)$, choose any x and put $e = y \diamond x = z \diamond x$. The two specializations of E677 give

$$x = y \diamond (x \diamond (e \diamond y)), \qquad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the first equality may be rewritten with left factor z. Left cancellation by z, then by x, and finally by e, gives $y = z$. Thus the labeling is injective. In particular, if $y \diamond x = z \diamond x = r$ with $y \neq z$, the two distinct permutations $\sigma(y), \sigma(z)$ agree at x but must separate at r:

$$\sigma(y)(r) \neq \sigma(z)(r), \qquad \sigma(\sigma(y)(r))(y) = r \backslash x = \sigma(\sigma(z)(r))(z).$$

The ETP affine-extension obstruction [4, Lemma 13.4] is useful calibration: linear or affine intuition should be checked against the right-collision rectangles before being used in this proof target.

8 Proof guardrails and trial-route notes

The following notes keep the proof package lean by marking which attractive shortcuts need an additional derivation before they can be used.

1. The family

   $$d_k \backslash x = c_{k-2} \diamond c_{k-1}$$

   is not part of the trusted package. The old derivation confused $c_{k-1} \diamond x$ with d_k; by definition it is d_{k-1}. A future proof may revive this family only by giving a new derivation.

2. The full-shift law

   $$p \diamond d_k = d_{k+1}$$

   is also not part of the trusted package and should be rederived before use.

3. A proof of universal idempotence, $d = 1$, or any universal small-period conclusion would contradict known nontrivial positive models from the ETP finite theory. Such a claim needs an explicit audit before it is used.

4. A proof that $H_x(t) = ((x \diamond t) \diamond x)$ has no nontrivial cycles is too strong as a global target: Remark 19 gives a positive $\mathbb{F}_5$ model with nontrivial H_x-cycles. The sharper target is fixed-point-free bad-point dynamics.

5. The ETP finite-map lemmas become useful after deriving a non-tautological self-map identity; the formal inverse identities for L_x and L_x^{-1} do not yet engage those lemmas.

6. A nontrivial collision $d_i = d_j$ can support a cascade proof only after the propagation $d_{i-1} = d_{j-1}$, or a replacement counting step, has been derived.

9 Proof targets

The work above leaves three operational proof targets. Target 12 records their common local bottleneck: prove Lemma L directly, or close the finite-counting splitter gap.

Target 24 (Fixed-point route)**.** Prove $q \diamond x = x$, equivalently Lemma L from Definition 15, by an argument that uses finite structure explicitly.

Target 25 (Collision route)**.** Resolve Target 22: from $d_i = d_j = e$, derive either the fixed point $q \diamond x = x$ or a finite counting contradiction for the injective splitter $A_e(y) = y \diamond e$ on the fiber $F_e = \{y : y \diamond x = e\}$.

Target 26 (Period-four route)**.** Eliminate the two remaining cases in Lemma 10: $r = p$, and $r \notin \{x, a, q, p\}$. This is the shortest current orbit-level proof target.

A Proof audit checklist

The reductions in this note use left cancellation only after Proposition 1, where finiteness has made each L_a bijective. They do not use right cancellation, associativity, an identity element, or group/quasigroup structure. Future work should state which proof target it attacks and end, if unfinished, with one next lemma rather than a broad search prescription.

Every new proof attempt should check the following list before being added to the mathematical record.

1. All products are parenthesized.

2. Every cancellation is a left cancellation by a known bijective left translation.

3. Every use of finiteness is named: left bijectivity, orbit periodicity, a finite-map lemma, or pigeonhole reasoning for a self-map.

4. The retired quotient family and full-shift law are used only if rederived.

5. Any claimed construction or structural constraint contains either a complete finite operation or a structural verification of the relevant identities.

References

[1] Matthew Bolan, Joachim Breitner, Jose Brox, Nicholas Carlini, Mario Carneiro, Floris van Doorn, Martin Dvorak, Andres Goens, Aaron Hill, Harald Husum, Hernan Ibarra Mejia, Zoltan A. Kocsis, Bruno Le Floch, Amir Livne Bar-on, Lorenzo Luccioli, Douglas McNeil, Alex Meiburg, Pietro Monticone, Pace P. Nielsen, Emmanuel Osalotioman Osazuwa, Giovanni Paolini, Marco Petracci, Bernhard Reinke, David Renshaw, Marcus Rossel, Cody Roux, Jeremy Scanvic, Shreyas Srinivas, Anand Rao Tadipatri, Terence Tao, Vlad Tsyrklevich, Fernando Vaquerizo-Villar, Daniel Weber, and Fan Zheng. The Equational Theories Project: Advancing collaborative mathematical research at scale. arXiv:2512.07087, 2025. https://arxiv.org/abs/2512.07087; DOI: https://doi.org/10.48550/arXiv.2512.07087.

[2] Equational Theories Project contributors. *The Equational Theories Project*. GitHub repository, 2024–2026. https://github.com/teorth/equational_theories.

[3] Equational Theories Project. Dashboard. https://teorth.github.io/equational_theories/dashboard/.

[4] Equational Theories Project. Blueprint, Chapter 13: Equation 677. https://teorth.github.io/equational_theories/blueprint/677-chapter.html.

[5] Equational Theories Project. Blueprint, Chapter 5: Finite implications and infinite models. https://teorth.github.io/equational_theories/blueprint/infinite-model-chapter.html.

[6] Equational Theories Project. Blueprint, Chapter 27: Hard implications. https://teorth.github.io/equational_theories/blueprint/hard-implications-chapter.html.

[7] A. K. Austin. A note on models of identities. *Proceedings of the American Mathematical Society*, 16:522–523, 1965. DOI: https://doi.org/10.1090/S0002-9939-1965-0174649-7; JSTOR: https://doi.org/10.2307/2034688.

[8] A. K. Austin. Finite models for laws in two variables. *Proceedings of the American Mathematical Society*, 17:1410–1412, 1966. DOI: https://doi.org/10.1090/S0002-9939-1966-0199134-9; JSTOR: https://doi.org/10.2307/2035753.

[9] Andrzej Kisielewicz. Varieties of algebras with no nontrivial finite members. In *Lattices, Semigroups, and Universal Algebra*, pages 129–136. Plenum, New York, 1990. DOI: https://doi.org/10.1007/978-1-4899-2608-1_14.

[10] A. Kisielewicz. Austin identities. *Algebra Universalis*, 38(3):324–328, 1997. DOI: https://doi.org/10.1007/s000120050057.

[11] Joel Berman and Stanley Burris. A computer study of 3-element groupoids. In *Logic and Algebra*, Lecture Notes in Pure and Applied Mathematics 180, pages 379–429. Marcel Dekker, 1996. Reprint DOI: https://doi.org/10.1201/9780203748671-18.

[12] Stanley Burris and H. P. Sankappanavar. *A Course in Universal Algebra*. Graduate Texts in Mathematics 78. Springer, New York, 1981. ISBN 978-0-387-90578-5.

[13] Donald E. Knuth. Notes on central groupoids. *Journal of Combinatorial Theory*, 8:376–390, 1970. DOI: https://doi.org/10.1016/S0021-9800(70)80032-1.

[14] Donald E. Knuth and Peter B. Bendix. Simple word problems in universal algebras. In *Computational Problems in Abstract Algebra*, pages 263–297. Pergamon, 1970. DOI: https://doi.org/10.1016/B978-0-08-012975-4.50028-X; reprint DOI: https://doi.org/10.1007/978-3-642-81955-1_23.

[15] Franz Baader and Tobias Nipkow. *Term Rewriting and All That*. Cambridge University Press, 1998. DOI: https://doi.org/10.1017/CBO9781139172752.

[16] Max Willsey, Chandrakana Nandi, Yisu Remy Wang, Oliver Flatt, Zachary Tatlock, and Pavel Panchekha. egg: Fast and extensible equality saturation. *Proceedings of the ACM on Programming Languages*, 5(POPL):1–29, 2021. DOI: https://doi.org/10.1145/3434304.

[17] Thomas Koehler, Andres Goens, Siddharth Bhat, Tobias Grosser, Phil Trinder, and Michel Steuwer. Guided equality saturation. *Proceedings of the ACM on Programming Languages*, 8(POPL):1727–1758, 2024. DOI: https://doi.org/10.1145/3632900.